

2_channel_Timer_IP USF Documentation

kunzelmann <robertniklas.kunzelmann@infineon.com>

September 8, 2026

Contents

1	2_channel_Timer_IP Introduction	5
2	2_channel_Timer_IP Block Diagram	6
3	2_channel_Timer_IP Primary Inputs	7
4	2_channel_Timer_IP Primary Outputs	10
5	2_channel_Timer_IP Registers	12
6	2_channel_Timer_IP Operations	15
7	2_channel_Timer_IP Operation Encoding	24
8	2_channel_Timer_IP Traces	31
8.1	sw_write_non_virtual_bf_TimEn_CH0_Trace Trace	31
8.2	hw_write_virtual_bf_ACTVAL_CH0_Trace Trace	31
8.3	read_CH0_CTRLSTAT_Trace Trace	31
8.4	Timer_CH0_dont_count_Trace Trace	32
8.5	Timer_CH0_reset_2_maxval_Trace Trace	32
8.6	Timer_CH0_down_count_Trace Trace	32
8.7	Timer_CH0_capture_CCUVAL0_Trace Trace	33
8.8	Timer_CH0_dont_capture_CCUVAL0_Trace Trace	33
8.9	Timer_CH0_compare_CCM1_CCU0_Trace Trace	34
8.10	Timer_CH0_compare_CCM5_CCU0_Trace Trace	34
8.11	Timer_CH0_disabled_compare_CCM7_CCU0_Trace Trace	34
8.12	Timer_CH0_overflow_int_reset_Trace Trace	35
8.13	Timer_CH0_overflow_int_set_Trace Trace	35

List of Figures

1	Top-level block diagram with primary inputs (left) and outputs (right).	6
2	Tabular overview (1) of the 2_channel_Timer_IP's operations encoding.	24
3	Tabular overview (2) of the 2_channel_Timer_IP's operations encoding.	25
4	Tabular overview (3) of the 2_channel_Timer_IP's operations encoding.	26
5	Tabular overview (4) of the 2_channel_Timer_IP's operations encoding.	27
6	Tabular overview (5) of the 2_channel_Timer_IP's operations encoding.	27
7	Tabular overview (6) of the 2_channel_Timer_IP's operations encoding.	28
8	Tabular overview (7) of the 2_channel_Timer_IP's operations encoding.	28
9	Tabular overview (8) of the 2_channel_Timer_IP's operations encoding.	29
10	Tabular overview (9) of the 2_channel_Timer_IP's operations encoding.	29
11	Tabular overview (10) of the 2_channel_Timer_IP's operations encoding.	30
12	Tabular overview (11) of the 2_channel_Timer_IP's operations encoding.	30
13	Tabular overview (12) of the 2_channel_Timer_IP's operations encoding.	30
14	Trace visualization as state chart.	31
15	Trace visualization as state chart.	31
16	Trace visualization as state chart.	32
17	Trace visualization as state chart.	32
18	Trace visualization as state chart.	33
19	Trace visualization as state chart.	33
20	Trace visualization as state chart.	33
21	Trace visualization as state chart.	34
22	Trace visualization as state chart.	34
23	Trace visualization as state chart.	35
24	Trace visualization as state chart.	35
25	Trace visualization as state chart.	35
26	Trace visualization as state chart.	36

List of Tables

1	List of the 2_channel_Timer_IP's primary inputs.	9
2	List of the 2_channel_Timer_IP's primary outputs.	11
3	List of the 2_channel_Timer_IP's registers.	14
4	List of the 2_channel_Timer_IP's operations.	23

1 2_channel_Timer_IP Introduction

Timer_IP contains two separately controlled 32-bit down counters sharing a rising-edge clock, active-high asynchronous reset, and a 32-bit register interface. CH0 has one capture/compare unit (CCU0); CH1 has six (CCU0 through CCU5), for seven CCUs total. There is no up-count mode, direction control, prescaler, channel chaining, or dedicated PWM/toggle state machine. ACTVAL is the software-readable and writable current count; MAXVAL is a software-programmed reload value, not an enforced upper bound. With TimEn=1, a reload condition has priority over counting. Otherwise, a nonzero ACTVAL decrements by one when CntIM=0 or the configured external-count condition is active. Zero does not wrap to 0xFFFFFFFF. ResIM=7 enables automatic reload from MAXVAL when ACTVAL is zero; modes 1 through 5 instead reload on the selected sampled external-reset condition, even before zero. ResIM=0 or 6 disables reload. All counter reloads and decrements require TimEn=1, but software writes, capture, comparison, and interrupt-condition evaluation do not. Global rst clears stored values and configuration to zero rather than loading MAXVAL. For CntIM, ResIM, and CapIM, modes 1/2/3/4/5 select sampled rising edge, falling edge, high level, low level, or either edge. CntIM=0 counts every clock; CntIM=6 or 7 supplies no count trigger. CapIM=0, 6, or 7 supplies no capture trigger. ResIM=7 is the special automatic-reload mode, not an external-input mode. The RTL input conditioners compare the current and preceding registered samples; level modes remain active each clock while the sampled level matches. External pins are not alternate timer clocks, and narrow asynchronous pulses are not guaranteed to be captured. CCM=0 captures ACTVAL into CCUVAL on the configured capture condition. CCM=1/2/3/4/5/6 drives the combinational unsigned comparison ACTVAL less than, less than or equal to, greater than, greater than or equal to, equal to, or unequal to CCUVAL. CCM=0 or 7 drives ExtComp low; 7 also disables hardware capture. Compare outputs are levels, not latched events or automatically generated pulses. ACTVAL and CCUVAL hardware writes take priority over simultaneous software writes to the same register. MAXVAL has no peripheral-write source. The term virtual bitfield denotes storage implemented in the timer/CCU rather than the control-register block, not an additional shadow copy. The signal named Ovflnt is a registered zero-condition output: on each clock it becomes 0 if OvflntRes is high, otherwise it becomes (ACTVAL=0 and ResIM!=0 and OvflntEn=1). It is not sticky, is not gated by TimEn, and can remain high at zero or clear without acknowledgement when the condition disappears. The default interface reads combinationally when rd_en=1 and writes selected registers on rising edges when wr_en=1; both may be asserted together. Ready is tied high in the RTL. Error is an ungated address-decode result, including during idle, not a general protocol checker. Exact word addresses are CH0 CTRLSTAT/ACTVAL/MAXVAL/CCUCTRL0/CCUVAL0 at 0x00/0x04/0x08/0x0C/0x10; CH1 CTRLSTAT/ACTVAL/MAXVAL at 0x14/0x18/0x1C and CCUCTRLn/CCUVALn at 0x20+8*n/0x24+8*n for n=0..5. CTRLSTAT contains TimEn[0], ResIM[3:1], CntIM[6:4], and OvflntEn[7], not a pending-interrupt status bit. CCUCTRLn contains CCMn[2:0] and CapIMn[5:3]. Unused read bits are zero, unused write bits are ignored, and no byte-enable or access-size input is present. The XML operations and traces describe selected behaviors, not one global implementation FSM. Important model limitations remain: all generated down-count traces exclude ACTVAL values 0 through 7, although the RTL decrements every nonzero value under the count condition; interrupt-set traces cover only ResIM=7 although RTL uses ResIM!=0, and interrupt-clear traces cover acknowledgement rather than all automatic deassertion cases. Ready has no defining action, interface-error traces cover invalid reads only, and virtual-register handshake/data helpers lack complete defining actions in this XML. These documentation corrections do not establish or repair full RTL/USF equivalence.

2 2_channel_Timer_IP Block Diagram

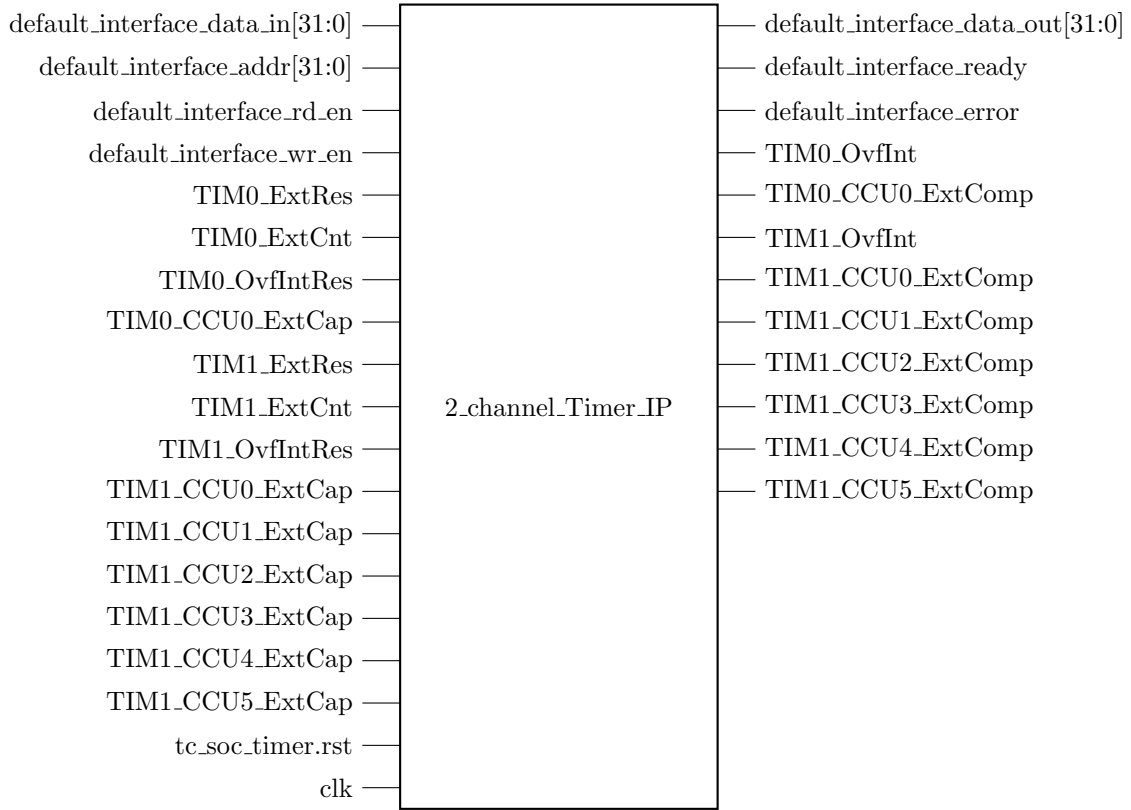


Figure 1: Top-level block diagram with primary inputs (left) and outputs (right).

3 2_channel_Timer_IP Primary Inputs

Input	Description
default_interface_data_in	The default_interface_data_in input carries the 32-bit software write word. With wr_en=1 and an exact mapped address, the selected register samples its implemented bits on the rising clock edge. CTRLSTAT uses bits [7:0], CCUC-TRL uses [5:0], and value registers use [31:0]; unused bits are ignored and there are no byte enables. A write to ACTVAL or CCUVAL takes effect only if its hardware-write enable is low, because hardware updates have priority. The source must meet clock setup/hold requirements; the constant-high ready output is not a write-completion pulse.
default_interface_addr	The default_interface_addr input is a 32-bit exact register address. The twenty mapped addresses are 0x00 through 0x4C in steps of four, with the channel/register allocation listed in the component description. The RTL compares all address bits: unaligned addresses and addresses with nonzero upper bits are not aliases and produce an address error. There is no transfer-size or byte-enable input. Read decoding is combinational; software write decoding additionally requires wr_en=1. The error output reflects whether this address is mapped even when no read or write is requested.
default_interface_rd_en	The default_interface_rd_en input enables the combinational read-data multiplexer. When high, a mapped address returns the current register word; when low, data_out is zero. An unmapped address also returns zero in the RTL. Reads do not clear status or change stored values. Read enable is independent of wr_en: when both are high the current value is readable while a write is eligible at the clock edge. The XML has separate read traces for wr_en=0 and wr_en=1; neither requires a multi-cycle read handshake.

default_interface_wr_en	The default_interface_wr_en input is the active-high software-write enable. An exact mapped address and wr_en=1 enable the addressed register's implemented fields at each rising clock edge, independently of rd_en. Holding wr_en high across clocks can repeat writes; there is no request queue or busy state. Hardware updates of ACTVAL and CCUVAL take priority over software writes to those registers, and still occur when wr_en=0 or another register is addressed. An unmapped address does not enable a software register write.
TIM0_ExtRes	The TIM0_ExtRes input is sampled by the channel's input conditioner and requests ACTVAL reload from MAXVAL according to ResIM: 1=rising edge, 2=falling edge, 3=high level, 4=low level, and 5=either edge. An active reload condition takes priority over counting but only writes ACTVAL when TimEn=1. Level modes can reload repeatedly while their sampled condition remains active. Modes 0, 6, and 7 ignore this pin; mode 7 instead reloads automatically at zero. This is not the global asynchronous rst input and does not clear configuration, captured values, or the interrupt register.
TIM0_ExtCnt	The TIM0_ExtCnt input is sampled on the common timer clock and supplies a count-enable condition for CntIM modes 1 through 5: rising edge, falling edge, high level, low level, or either edge. It is not an alternate clock. Counting still requires TimEn=1, a nonzero ACTVAL, and no reload condition. High/low level modes count clock cycles at the selected level, not just external edges. Mode 0 ignores this pin and counts every clock; modes 6 and 7 provide no count trigger. Events shorter than the sampling interval may be missed.

TIM0_OvfIntRes	The TIM0_OvfIntRes input forces TIM0_OvfInt to zero on a rising clock edge and has priority over the interrupt condition. It is connected directly to the interrupt register's synchronous-clear logic, without the configurable input conditioner used for ExtRes/ExtCnt/ExtCap, so the source must satisfy the timer clock's timing requirements. It does not reload ACTVAL or clear configuration. OvfInt is not sticky: it also clears when its condition disappears and can reassert on the next clock after OvfIntRes is released if ACTVAL remains zero, ResIM is nonzero, and OvfIntEn is set.
TIM0_CCU0_ExtCap	The TIM0_CCU0_ExtCap input is sampled and interpreted according to CapIM0. Only in CCM0=0 does the resulting condition cause CCUVAL0 to capture the pre-update ACTVAL value on a rising clock edge. Modes 1/2/3/4/5 select sampled rising edge, falling edge, high level, low level, or either edge; modes 0/6/7 disable the trigger. Level modes may capture on consecutive clocks. Capture does not require TimEn and overrides a simultaneous software write to CCUVAL0. The stored value corresponds to the clocked, conditioned event, not the exact asynchronous pin-transition time; narrow pulses may be missed.

Table 1: List of the 2_channel_Timer_IP's primary inputs.

4 2_channel_Timer_IP Primary Outputs

Output	Description
default_interface_data_out	The default_interface_data_out output is a 32-bit combinational read result, not a registered response. With rd.en=1 and a mapped address it reflects the current register value, with unused CTRLSTAT/CCUCTRL bits zero-filled. It is zero when rd.en=0 or the address is unmapped in the RTL. Simultaneous reading and writing is not prohibited: readback changes as the underlying register updates at the clock edge. The XML default-data operation covers rd.en=0; its invalid-read traces assert error without also constraining data_out to zero, so that RTL default is not fully specified by those traces.
default_interface_ready	The default_interface_ready output is tied to 1 in the RTL, including during reset and when no request is active. It does not pulse on completion, indicate a pending transaction, or suppress address errors. Reads are combinational and enabled writes occur on clock edges. The XML declares this non-register output with ResetValue=0 but supplies no defining operation or trace action for it; that metadata does not describe the RTL constant-high behavior. This model gap is not corrected by changing the description.
default_interface_error	The default_interface_error output is 1 whenever default_interface_addr differs from all twenty exact mapped word addresses and is 0 for a mapped address. In the RTL this is a direct combinational address-decode result, independent of rd.en, wr.en, and ready; it can therefore be high during idle or write-only cycles. It does not detect invalid configuration values, simultaneous reads/writes, or general protocol errors. The XML interface-error traces constrain only unmapped reads and contain no matching complete low/default behavior, so they cover less than the RTL function.

TIM0_OvfInt	The TIM0_OvfInt output resets asynchronously to zero. At each rising clock edge it becomes zero if TIM0_OvfIntRes=1; otherwise it becomes (ACTVAL=0 and ResIM!=0 and OvfIntEn=1), evaluated from the pre-edge values. It indicates the down counter's zero condition, not exceeding MAXVAL or arithmetic wraparound. TimEn is not part of this condition. The output can be a pulse during automatic reload, stay high while an enabled condition persists at zero, or clear automatically when the condition disappears. The XML set trace covers ResIM=7 only, and its clear trace covers OvfIntRes=1 rather than all RTL deassertion cases.
TIM0_CCU0_ExtComp	The TIM0_CCU0_ExtComp output is the combinational unsigned comparison selected by CCM0 between ACTVAL and CCUVAL0. Modes 1 through 6 select less than, less than or equal to, greater than, greater than or equal to, equal to, and unequal to, respectively; the output is 1 when that relation is true and 0 when false. Modes 0 and 7 force it low. It is not equality-only, not latched, and not automatically a one-clock pulse. Comparison remains active when TimEn=0 and can change when software writes either compared value or CCM0.

Table 2: List of the 2_channel_Timer IP's primary outputs.

5 2_channel_Timer_IP Registers

Register	Description
bf_TimEn_CH0_out	The bf_TimEn_CH0_out object is CH0_CTRL-STAT bit 0, a one-bit internal configuration register reset to zero. When set, it allows the ACTVAL hardware control logic to decrement or reload the counter according to CntIM and ResIM. Clearing it suppresses those hardware writes and preserves ACTVAL unless software writes it. It does not disable software register access, capture, comparison, input sampling, or overflow-condition evaluation; those functions have no TimEn gate in the RTL.
bf_ResIM_CH0_out	The bf_ResIM_CH0_out object is the three-bit encoded reset-input mode in CH0_CTRL-STAT[3:1], reset to 0; the bits are not independent masks. Values 1, 2, 3, 4, and 5 select sampled rising edge, falling edge, high level, low level, and either edge of TIM0_ExtRes. Value 7 selects automatic reload when ACTVAL=0. Values 0 and 6 generate no reload condition. Reload writes MAXVAL to ACTVAL only when TimEn=1 and takes priority over decrementing; an external reload is not restricted to zero. Separately, any nonzero ResIM enables the zero-condition term used by the RTL interrupt logic, including mode 6. This differs from the XML interrupt-set traces, which cover only ResIM=7.
bf_CntIM_CH0_out	The bf_CntIM_CH0_out object is CH0_CTRL-STAT[6:4], a three-bit encoded count-input mode reset to 0. Mode 0 allows a decrement on every timer clock. Modes 1, 2, 3, 4, and 5 select sampled rising edge, falling edge, high level, low level, and either edge of TIM0_ExtCnt. Modes 6 and 7 supply no count trigger. A decrement additionally requires TimEn=1, ACTVAL!=0, and no reload condition. Level modes can decrement on every clock while the sampled level matches; edge modes react to changes between registered samples. This field selects a count enable, not a separate clock or a prescaler.

bf_OvfIntEn_CH0_out	The bf_OvfIntEn_CH0_out object is CH0_CTRLSTAT bit 7, reset to zero. It enables the registered zero-condition indication TIM0_OvfInt; it is not a pending-status flag. On each clock, unless TIM0_OvfIntRes clears the output, the RTL loads (ACTVAL=0 and ResIM!=0 and OvfIntEn=1) into TIM0_OvfInt. The condition does not require TimEn, and clearing OvfIntEn causes deassertion at the subsequent evaluating clock rather than preserving a pending interrupt. XML interrupt-set traces describe only the ResIM=7 subset.
bf_ACTVAL_CH0_flipflop_out	The bf_ACTVAL_CH0_flipflop_out object is CH0's 32-bit current countdown register, readable and writable at address 0x04 and reset to zero. The RTL decrements it under the enabled count condition, reloads it from MAXVAL under the enabled reset condition, and otherwise retains it unless software writes it. Hardware decrement or reload has priority over a simultaneous software write to ACTVAL. Zero is held until reload or software initialization; MAXVAL does not clamp a software-written count. Storage resides in the timer channel, while the register-interface block supplies its write data and enable. The XML down-count guards omit values 1 through 7 even though those values count normally in the RTL.
bf_MAXVAL_CH0_flipflop_out	The bf_MAXVAL_CH0_flipflop_out object is the 32-bit reload register at address 0x08, reset to zero. Software writes it; there is no hardware-write source for MAXVAL. The counter loads its current value only on an enabled external reload or on automatic reload at zero with ResIM=7. Writing MAXVAL alone does not initialize ACTVAL, enforce an upper bound, or select an up-count mode. With continuous internal counting, TimEn=1, and automatic reload, successive reloads are MAXVAL+1 clocks apart because zero occupies a cycle; external count/reload conditions alter that timing. Global rst clears MAXVAL and ACTVAL rather than performing this reload.

bf_CCM0_CH0_out	The bf_CCM0_CH0_out object is CCUC-TRL0[2:0] at address 0x0C and controls CCU0 only. It resets to 0, selecting capture on the condition configured by CapIM0. Modes 1/2/3/4/5/6 drive ExtComp with the unsigned comparison ACTVAL less than, less than or equal to, greater than, greater than or equal to, equal to, or unequal to CCUVAL0. Mode 7 disables hardware capture and drives ExtComp low; capture mode 0 also drives ExtComp low. Compare outputs are combinational Boolean levels, not latched events. Neither capture nor compare requires TimEn=1, and software can write CCUVAL0 in any mode when no hardware capture takes priority.
bf_CapIM0_CH0_out	The bf_CapIM0_CH0_out object is CCUC-TRL0[5:3] at address 0x0C, reset to 0. It is an encoded input mode: 1=rising edge, 2=falling edge, 3=high level, 4=low level, and 5=either edge of the sampled TIM0_CCU0_ExtCap input. Values 0, 6, and 7 produce no capture trigger. The condition writes ACTVAL into CCUVAL0 only when CCM0=0. Level modes can overwrite the capture value on consecutive clocks; there is no one-shot latch or capture FIFO. The RTL uses two sample-history registers and observes the first sample for level detection, so this field must not be described as guaranteeing two-stage CDC filtering or capture of arbitrarily narrow pulses.
bf_CCUVAL0_CH0_flipflop_out	The bf_CCUVAL0_CH0_flipflop_out object is CCU0's 32-bit capture/compare register at address 0x10, reset to zero. In capture mode (CCM0=0), the configured sampled capture condition writes the pre-update ACTVAL value on a clock edge, independently of TimEn. Otherwise it retains its value unless software writes it. A hardware capture takes priority over a simultaneous software write. In modes 1 through 6 the stored value is the unsigned compare reference, not a capture-event threshold. This is one register, with no capture queue or extra shadow storage.

Table 3: List of the 2_channel_Timer_IP's registers.

6 2_channel_Timer_IP Operations

Operations	Description
read_CH0_CTRLSTAT	The read_CH0_CTRLSTAT operation supplies the control word at address 0x00: TimEn[0], ResIM[3:1], CntIM[6:4], and OvflntEn[7]. The RTL zero-fills bits [31:8]. Despite the CTRLSTAT name, the word contains no pending-overflow status bit. Readback is combinational with rd_en=1 for either value of wr_en and has no read side effects. The operation specifies the concatenated fields; the associated traces select the address and enable conditions.
read_CH0_ACTVAL	The read_CH0_ACTVAL operation returns the current 32-bit ACTVAL for an enabled read at address 0x04. In the RTL, bf_ACTVAL_CH0_out connects directly to the channel's ACTVAL storage, not a separate snapshot. Reading neither stops nor clears the counter; the result can change at a clock edge due to a hardware or software update. ACTVAL is also software-writable, subject to hardware-write priority.
read_CH0_MAXVAL	The read_CH0_MAXVAL operation returns the programmed 32-bit reload value for rd_en=1 at address 0x08. MAXVAL is loaded into ACTVAL only on an enabled external reload or automatic reload with ResIM=7. It is not an enforced count limit and does not by itself determine a time period: count-input mode, timer enable, and reload conditions also matter. This combinational read modifies neither register.
read_CH0_CCUCTRL0	The read_CH0_CCUCTRL0 operation returns CCM0 in bits [2:0] and CapIM0 in [5:3] for rd_en=1 at address 0x0C; RTL bits [31:6] are zero-filled. There is no separate CCU enable or status bit. CCM0 selects capture, a comparison relation, or disabled mode, while CapIM0 selects the sampled capture condition. Readback is combinational and has no side effects.

read_CH0_CCUVAL0	The read_CH0_CCUVAL0 operation returns the current 32-bit CCUVAL0 for rd_en=1 at address 0x10. This is the most recent hardware capture or software-written value and serves as the reference in compare modes. Readback is combinational, does not consume a capture or acknowledge an event, and does not prevent a later capture from overwriting the single stored value.
default_data_out_default_interface	The default_data_out_default_interface operation assigns the literal value zero to default_interface_data_out; it does not return a selected register's contents. The XML selects this zero-length behavior when rd_en=0, with separate traces for wr_en=0 and wr_en=1. The RTL also returns zero for an unmapped address while rd_en=1, but that case is not covered by this operation's generated traces.
interface_error_default_interface	The interface_error_default_interface operation assigns default_interface_error=1. Its zero-length XML traces require rd_en=1 and an address unequal to every mapped word address, for either value of wr_en. This is not a software error report or a general protocol check. The RTL error output is high for an unmapped address even during idle or a write-only cycle and low for mapped addresses. The XML does not fully specify those cases or the zero read-data result on an invalid read.
sw_write_non_virtual_bf_TimEn_CH0	The sw_write_non_virtual_bf_TimEn_CH0 operation loads TimEn from previous-cycle default_interface_data_in[0], modeling the clocked update after wr_en=1 at address 0x00. The same word write also updates ResIM, CntIM, and OvflntEn; these fields are not separately addressed. TimEn gates hardware counter decrement and reload only, not capture, comparison, interrupt-condition evaluation, or software access. The new bit takes effect after the writing clock edge, not asynchronously when write data changes.

sw_write_non_virtual_bf_ResIM_CH0	The sw_write_non_virtual_bf_ResIM_CH0 operation loads ResIM from previous-cycle write-data bits [3:1] after an enabled write at address 0x00. It updates the encoded mode on a clock edge together with the other CTRLSTAT fields. Values 1 through 5 select external sampled rising/falling/high/low/either-edge reload conditions, 7 selects automatic reload at zero, and 0 or 6 disables reload. Reload also requires TimEn=1. The RTL interrupt zero-condition uses ResIM!=0 independently of the reload mode's effect.
sw_write_non_virtual_bf_CntIM_CH0	The sw_write_non_virtual_bf_CntIM_CH0 operation loads CntIM from previous-cycle write-data bits [6:4] after wr_en=1 at address 0x00. It is a clocked update of the count-enable mode, not a clock-source switch. Mode 0 counts every timer clock, modes 1 through 5 select sampled external edge/level conditions, and modes 6 and 7 disable the count trigger. TimEn, nonzero ACTVAL, and reload priority still govern whether a decrement occurs.
sw_write_non_virtual_bf_OvfIntEn_CH0	The sw_write_non_virtual_bf_OvfIntEn_CH0 operation loads OvfIntEn from previous-cycle write-data bit 7 after a write at address 0x00. It enables the registered zero-condition indication; it does not directly write or acknowledge TIM0_OvfInt. The interrupt register evaluates the pre-edge enable value together with ACTVAL=0, ResIM!=0, and TIM0_OvfIntRes. There is no software-writable pending-interrupt field in CTRLSTAT.
sw_write_virtual_bf_ACTVAL_CH0	The sw_write_virtual_bf_ACTVAL_CH0 operation loads ACTVAL storage from previous-cycle 32-bit default_interface_data_in. Its XML selection requires wr_en=1, address=0x04, and bf_ACTVAL_CH0_peripheral_wr_en=0. The RTL gives hardware decrement or reload priority over a simultaneous software write; constant-high ready does not mean the software value won that conflict. Software can set TimEn=0 before initializing ACTVAL to suppress hardware counter writes. No MAXVAL clamp is applied.

sw_write_virtual_bf_MAXVAL_CH0	The sw_write_virtual_bf_MAXVAL_CH0 operation loads MAXVAL with previous-cycle 32-bit write data after wr_en=1 at address 0x08. MAXVAL storage resides in the timer channel but has no peripheral-write source. Writing it does not immediately change ACTVAL or start the timer. A later enabled external or automatic reload uses MAXVAL; if MAXVAL is written on the same edge as a reload, the RTL reload uses the pre-edge MAXVAL value.
sw_write_non_virtual_bf_CCM0_CH0	The sw_write_non_virtual_bf_CCM0_CH0 operation loads CCM0 from previous-cycle write-data bits [2:0] after a write at address 0x0C; that word also updates CapIM0. Mode 0 selects capture, 1 through 6 select unsigned comparison relations, and 7 disables hardware capture and the compare output. CCM0 is a mode field for CCU0, not a compare-value register. Its update is clocked, while the resulting compare output is combinational and is not gated by TimEn.
sw_write_non_virtual_bf_CapIM0_CH0	The sw_write_non_virtual_bf_CapIM0_CH0 operation loads CapIM0 from previous-cycle write-data bits [5:3] after a write at address 0x0C, together with CCM0. Modes 1/2/3/4/5 select sampled rising edge, falling edge, high level, low level, and either edge of TIM0_CCU0.ExtCap; 0/6/7 supply no capture trigger. This configures a condition rather than issuing a capture command. Hardware capture requires CCM0=0. Changing CapIM0 does not reset the RTL input-sample history.
sw_write_virtual_bf_CCUVAL0_CH0	The sw_write_virtual_bf_CCUVAL0_CH0 operation loads CCUVAL0 with previous-cycle 32-bit write data when wr_en=1, address=0x10, and its peripheral-write enable is low. A simultaneous hardware capture wins over the software write in the RTL and XML selection. In compare modes the stored value is the comparison reference; in capture mode it is simply the value held until a later capture or software write. It does not set a threshold for detecting an external capture event.

hw_write_virtual_bf_ACTVAL_CH0	The hw_write_virtual_bf_ACTVAL_CH0 operation assigns ACTVAL storage from previous-cycle bf_ACTVAL_CH0_flipflop_in when the peripheral-write enable is high. In the RTL this selected input is MAXVAL for an enabled reload or ACTVAL-1 for an enabled decrement. Hardware writes have priority regardless of wr_en or the software address. This is an automatic update of the same software-visible storage, not a software-invoked synchronization command. The XML references the peripheral-write and selected-input helpers without fully defining their combinational equations; direct counter traces describe further update cases separately.
hw_write_virtual_bf_CCUVAL0_CH0	The hw_write_virtual_bf_CCUVAL0_CH0 operation assigns CCUVAL0 storage from previous-cycle bf_CCUVAL0_CH0_flipflop_in when its peripheral-write enable is high. The RTL hardware source is pre-update ACTVAL during a configured capture in CCM0=0, not a programmable compare-threshold update. Capture is independent of TimEn and overrides a simultaneous software write. This is a formal description of automatic hardware behavior, not a software API; its selected-input and enable helpers lack complete defining actions in this XML.
reg_hold_bf_ACTVAL_CH0	The reg_hold_bf_ACTVAL_CH0 operation preserves the previous-cycle ACTVAL storage value. Its selection requires no peripheral write and no software write to address 0x04, including cases where software writes another register. This represents an inactive storage write enable in the RTL, not a separate hold-control port. It must not be applied while a hardware decrement/reload or an accepted software ACTVAL write is active.
reg_hold_bf_CCUVAL0_CH0	The reg_hold_bf_CCUVAL0_CH0 operation preserves the previous-cycle CCUVAL0 storage value when no peripheral capture write and no software write to address 0x10 is selected. The value remains available for reads and comparison. This is a formal hold case, not a command that blocks other writes; an active capture or accepted software write instead updates the register.

Timer_CH0_down_count	The Timer_CH0_down_count operation specifies ACTVAL as previous-cycle ACTVAL minus one. It describes automatic hardware counting, not a software command. The RTL updates when TimEn=1, ACTVAL!=0, no reload is requested, and either CntIM=0 or the selected sampled external-count condition is active. It neither implements up-counting nor decrements zero into wraparound. All generated CH0 down-count trace guards exclude values 0 through 7 rather than just zero, so they omit RTL decrements from values 1 through 7.
Timer_CH0_dont_count	The Timer_CH0_dont_count operation specifies ACTVAL equal to its previous-cycle stored value. Its traces select particular disabled, zero, or inactive-trigger cases and exclude the ACTVAL address for possible software access. It is not a blanket prohibition on updating ACTVAL: an enabled reload or accepted software write can change the counter when no decrement occurs. The RTL supports countdown only; this hold operation does not imply an increment mode.
Timer_CH0_capture_CCUVAL0	The Timer_CH0_capture_CCUVAL0 operation specifies CCUVAL0 equal to previous-cycle ACTVAL and constrains previous-cycle ExtComp to zero for capture mode. Its traces require CCM0=0 and the selected sampled capture condition. In the RTL the register captures pre-update ACTVAL on a clock edge, even if counting or reload changes ACTVAL on the same edge. Capture is independent of TimEn, has priority over a software CCUVAL0 write, and may repeat in level modes. It is not a timestamp of the exact asynchronous input-transition time.
Timer_CH0_dont_capture_CCUVAL0	The Timer_CH0_dont_capture_CCUVAL0 operation preserves previous-cycle CCUVAL0 and constrains previous-cycle ExtComp to zero in capture mode. Its traces cover selected inactive capture conditions and exclude address 0x10 to avoid the software-write case. It does not prevent software from writing CCUVAL0 when capture is inactive, and CCUVAL0 is the capture register rather than the counter itself. Holding the register requires the absence of both hardware capture and an accepted software update.

Timer_CH0_compare_CCM1_CCU0	With CCM0=1, Timer_CH0_compare_CCM1_CCU0 drives TIM0_CCU0_ExtComp to the unsigned relation ACTVAL less than CCUVAL0: 1 when true and 0 otherwise. It is a zero-cycle combinational comparison of current stored values, not an edge-triggered or latched event, and does not require TimEn=1.
Timer_CH0_compare_CCM2_CCU0	With CCM0=2, Timer_CH0_compare_CCM2_CCU0 drives TIM0_CCU0_ExtComp to the unsigned relation ACTVAL less than or equal to CCUVAL0: 1 when true and 0 otherwise. The comparison is combinational, independent of TimEn, and can remain high for multiple clocks. It does not itself generate a pulse, latch an event, or modify either register.
Timer_CH0_compare_CCM3_CCU0	With CCM0=3, Timer_CH0_compare_CCM3_CCU0 drives TIM0_CCU0_ExtComp to the unsigned relation ACTVAL greater than CCUVAL0: 1 when true and 0 otherwise. This describes a combinational Boolean level, not a timer-state transition or latched interrupt. It remains applicable when TimEn=0.
Timer_CH0_compare_CCM4_CCU0	With CCM0=4, Timer_CH0_compare_CCM4_CCU0 drives TIM0_CCU0_ExtComp to the unsigned relation ACTVAL greater than or equal to CCUVAL0: 1 when true and 0 otherwise. The combinational result depends on current stored values and does not require an enabled or running counter. Neither register is changed by the operation.
Timer_CH0_compare_CCM5_CCU0	With CCM0=5, Timer_CH0_compare_CCM5_CCU0 drives TIM0_CCU0_ExtComp to ACTVAL equal to CCUVAL0: 1 when equal and 0 otherwise. The output is combinational and can remain high indefinitely if equal values are held, including while TimEn=0. Equality can also result from software writes or reload; it does not prove that the counter crossed a threshold or that a new event occurred.

Timer_CH0_compare_CCM6_CCU0	With CCM0=6, Timer_CH0_compare_CCM6_CCU0 drives TIM0_CCU0_ExtComp to ACTVAL unequal to CCUVAL0: 1 when different and 0 when equal. It is a combinational level based on current stored values, independent of TimEn, not an error indication or a latched mismatch event.
Timer_CH0_disabled_compare_CCM7_CCU0	The Timer_CH0_disabled_compare_CCM7_CCU0 operation drives TIM0_CCU0_ExtComp=0 when CCM0 equals 7. It applies regardless of TimEn and does not disable the counter or software access to CCUVAL0. The RTL also suppresses hardware capture in mode 7. Capture mode 0 separately forces ExtComp low, but is not this operation's trigger.
Timer_CH0_reset_2_maxval	The Timer_CH0_reset_2_maxval operation assigns ACTVAL the previous-cycle MAXVAL. In the RTL reload requires TimEn=1 and either the sampled external-reset condition for ResIM=1..5 or ACTVAL=0 with ResIM=7. Reload takes priority over decrement and a simultaneous software ACTVAL write. External reload can occur at nonzero count and is independent of the count trigger. This is not global reset and does not clear configuration, CCUVAL, or the interrupt register. Only down-counting is implemented; MAXVAL is not an enforced upper bound.
Timer_CH0_overflow_int_set	The Timer_CH0_overflow_int_set operation assigns TIM0_OvfInt=1. Its length-one XML trace requires OvfIntRes=0, OvfIntEn=1, ACTVAL=0, and ResIM=7; TimEn is not tested. It describes a hardware register update, not a manual software command or writable interrupt flag. The RTL asserts for any nonzero ResIM under the other conditions, so this trace covers only a subset. OvfInt is recomputed every clock rather than held until acknowledged.

Timer_CH0_overflow_int_reset	The Timer_CH0_overflow_int_reset operation assigns TIM0_OvfInt=0 for the trace case TIM0_OvfIntRes=1. The RTL samples this dedicated input on the clock edge with priority over the set condition; there is no software-addressed interrupt-clear register or write-one-to-clear field. The RTL also clears on global reset or when ACTVAL is nonzero, ResIM=0, or OvfIntEn=0. Those automatic deassertion cases are not all represented by this acknowledgement-only trace.
------------------------------	---

Table 4: List of the 2_channel_Timer_IP's operations.

7 2_channel_Timer_IP Operation Encoding

[illegible]

Figure 2: Tabular overview (1) of the 2_channel_Timer_IP’s operations encoding.

bf_TimEn_CHO_cnt	default_interface_addr	bf_ACTUAL_CHO_flagdp_cnt	bf_ResM_CHO_cnt	Tim0_ExtRes	bf_CntM_CHO_cnt	Tim0_ExtCnt	Operation
0	-	-	7	-	-	-	Timer_CHO_reset_2_maxvsa
	-	-	3	1	-	-	Timer_CHO_reset_2_maxvsa
	-	-	0	-	-	-	Timer_CHO_lont_count
	-	-	4	1	-	-	Timer_CHO_reset_2_maxvsa
	-	-	6	-	-	-	Timer_CHO_lont_count
	-	-	0	-	-	-	Timer_CHO_lont_count
	-	-	1	-	-	-	Timer_CHO_reset_2_maxvsa
	-	-	2	-	-	-	Timer_CHO_lont_count
	-	-	5	-	-	-	Timer_CHO_reset_2_maxvsa
	-	-	1	-	-	-	Timer_CHO_reset_2_maxvsa
	-	-	3	1	-	-	Timer_CHO_lont_count
	-	-	4	0	-	-	Timer_CHO_lont_count
	-	-	6	-	-	-	Timer_CHO_lont_count
	-	-	0	-	-	-	Timer_CHO_lont_count
	-	-	1	-	-	-	Timer_CHO_lont_count
	-	-	2	-	-	-	Timer_CHO_lont_count
	-	-	5	-	-	-	Timer_CHO_lont_count
	-	-	3	0	-	-	Timer_CHO_lont_count
	-	-	4	1	-	-	Timer_CHO_lont_count
	-	-	6	-	-	-	Timer_CHO_lont_count
	-	-	0	-	-	-	Timer_CHO_lont_count
	-	-	1	-	-	-	Timer_CHO_lont_count
	-	-	2	-	-	-	Timer_CHO_lont_count
	-	-	5	-	-	-	Timer_CHO_lont_count
	-	-	3	1	-	-	Timer_CHO_lont_count
	-	-	4	0	-	-	Timer_CHO_lont_count
	-	-	6	-	-	-	Timer_CHO_lont_count
	-	-	0	-	-	-	Timer_CHO_lont_count
	-	-	1	-	-	-	Timer_CHO_lont_count
	-	-	2	-	-	-	Timer_CHO_lont_count
	-	-	5	-	-	-	Timer_CHO_lont_count
	-	-	3	1	-	-	Timer_CHO_lont_count
	-	-	4	0	-	-	Timer_CHO_lont_count
	-	-	6	-	-	-	Timer_CHO_lont_count
	-	-	0	-	-	-	Timer_CHO_lont_count
	-	-	1	-	-	-	Timer_CHO_lont_count
	-	-	2	-	-	-	Timer_CHO_lont_count
	-	-	5	-	-	-	Timer_CHO_lont_count
	-	-	3	1	-	-	Timer_CHO_lont_count
	-	-	4	0	-	-	Timer_CHO_lont_count
	-	-	6	-	-	-	Timer_CHO_lont_count
	-	-	0	-	-	-	Timer_CHO_lont_count
	-	-	1	-	-	-	Timer_CHO_lont_count
	-	-	2	-	-	-	Timer_CHO_lont_count
	-	-	5	-	-	-	Timer_CHO_lont_count
	-	-	3	1	-	-	Timer_CHO_lont_count
	-	-	4	0	-	-	Timer_CHO_lont_count
	-	-	6	-	-	-	Timer_CHO_lont_count
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	
-	-	6	-	-	-	Timer_CHO_lont_count	
-	-	0	-	-	-	Timer_CHO_lont_count	
-	-	1	-	-	-	Timer_CHO_lont_count	
-	-	2	-	-	-	Timer_CHO_lont_count	
-	-	5	-	-	-	Timer_CHO_lont_count	
-	-	3	1	-	-	Timer_CHO_lont_count	
-	-	4	0	-	-	Timer_CHO_lont_count	

Figure 3: Tabular overview (2) of the 2_channel_Timer_IP’s operations encoding.

bf_TimEn_CH1_out	default_interface_addr	bf_ACTVAL_CH1_dlylip_out	bf_ResM_CH1_out	TIM1_ExtRes	bf_CntM_CH1_out	TIM1_ExtCnt	Operation
0	-	-	-	-	-	-	Timer_CH1_zlont_count
-	-	-	7	-	-	-	Timer_CH1_reset_2_maxvva
-	-	-	3	1	-	-	Timer_CH1_zlont_count
-	-	-	-	0	-	-	Timer_CH1_zlont_count
-	-	-	4	-	-	-	Timer_CH1_reset_2_maxvva
-	-	-	-	1	-	-	Timer_CH1_zlont_count
-	-	-	6	-	-	-	Timer_CH1_zlont_count
-	-	0	0	-	-	-	Timer_CH1_zlont_count
-	-	-	1	-	-	-	Timer_CH1_reset_2_maxvva
-	-	-	-	-	-	-	Timer_CH1_zlont_count
-	-	-	2	-	-	-	Timer_CH1_reset_2_maxvva
-	-	-	-	-	-	-	Timer_CH1_zlont_count
-	-	-	-	-	-	-	Timer_CH1_reset_2_maxvva
-	-	-	5	-	-	-	Timer_CH1_zlont_count
-	-	-	-	1	-	-	Timer_CH1_reset_2_maxvva
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	-	0	Timer_CH1_zlont_count
-	-	-	-	-	4	1	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	3	0	1	-	Timer_CH1_down_count
-	-	-	-	-	-	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_down_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	-	-	Timer_CH1_reset_2_maxvva
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	-	0	Timer_CH1_zlont_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_down_count
-	-	-	-	-	5	-	Timer_CH1_down_count
-	-	-	-	-	3	1	Timer_CH1_zlont_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1_zlont_count
-	-	-	-	-	6	-	Timer_CH1_zlont_count
-	-	-	-	-	7	-	Timer_CH1_zlont_count
-	-	-	-	-	0	-	Timer_CH1_zlont_count
-	-	-	-	-	1	-	Timer_CH1_zlont_count
-	-	-	-	-	2	-	Timer_CH1_zlont_count
-	-	-	-	-	5	-	Timer_CH1_zlont_count
-	-	-	-	-	3	1	Timer_CH1_down_count
-	-	-	-	-	4	-	Timer_CH1

Figure 4: Tabular overview (3) of the 2_channel_Timer_IP’s operations encoding.

bf_CCM0.CH0_out	bf_CapIM0.CH0_out	TIM0.CCU0.ExtCap	default_interface_addr	Operation
0	3	1	-	Timer.CH0.capture_CCUVAL0
		0	-	Timer.CH0.dont_capture_CCUVAL0
	4	1	-	Timer.CH0.capture_CCUVAL0
			-	Timer.CH0.dont_capture_CCUVAL0
	6	-	Timer.CH0.dont_capture_CCUVAL0	
	7	-	Timer.CH0.dont_capture_CCUVAL0	
	0	-	Timer.CH0.dont_capture_CCUVAL0	
	1	-	-	Timer.CH0.capture_CCUVAL0
			-	Timer.CH0.dont_capture_CCUVAL0
	2		-	Timer.CH0.capture_CCUVAL0
			-	Timer.CH0.dont_capture_CCUVAL0
	5		-	Timer.CH0.capture_CCUVAL0
1	-	-	Timer.CH0.compare_CCM1.CCU0	
2	-	-	Timer.CH0.compare_CCM2.CCU0	
3	-	-	Timer.CH0.compare_CCM3.CCU0	
4	-	-	Timer.CH0.compare_CCM4.CCU0	
5	-	-	Timer.CH0.compare_CCM5.CCU0	
6	-	-	Timer.CH0.compare_CCM6.CCU0	
7	-	-	Timer.CH0.disabled_compare_CCM7.CCU0	

Figure 5: Tabular overview (4) of the 2_channel_Timer_IP's operations encoding.

bf_CCM0_CH1_out	bf_CapIM0_CH1_out	TIM1_CCU0_ExtCap	default_interface_addr	Operation
0	3	1	-	Timer_CH1.capture_CCUVAL0
		0	-	Timer_CH1.dont.capture_CCUVAL0
	4	1	-	Timer_CH1.capture_CCUVAL0
			-	Timer_CH1.dont.capture_CCUVAL0
	6	-	Timer_CH1.dont.capture_CCUVAL0	
	7	-	Timer_CH1.dont.capture_CCUVAL0	
	0	-	Timer_CH1.dont.capture_CCUVAL0	
	1	-	-	Timer_CH1.capture_CCUVAL0
			-	Timer_CH1.dont.capture_CCUVAL0
	2		-	Timer_CH1.capture_CCUVAL0
			-	Timer_CH1.dont.capture_CCUVAL0
	5		-	Timer_CH1.capture_CCUVAL0
1	-	-	Timer_CH1.compare_CCM1_CCU0	
2	-	-	Timer_CH1.compare_CCM2_CCU0	
3	-	-	Timer_CH1.compare_CCM3_CCU0	
4	-	-	Timer_CH1.compare_CCM4_CCU0	
5	-	-	Timer_CH1.compare_CCM5_CCU0	
6	-	-	Timer_CH1.compare_CCM6_CCU0	
7	-	-	Timer_CH1.disabled.compare_CCM7_CCU0	

Figure 6: Tabular overview (5) of the 2_channel_Timer_IP's operations encoding.

bf_CCM1.CH1_out	bf_CapIM1.CH1_out	TIM1.CCU1.ExtCap	default_interface_addr	Operation
0	3	1	-	Timer.CH1.capture_CCUVAL1
		0	-	Timer.CH1.dont_capture_CCUVAL1
	4	1	-	Timer.CH1.capture_CCUVAL1
			-	Timer.CH1.dont_capture_CCUVAL1
	6	-	Timer.CH1.dont_capture_CCUVAL1	
	7	-	Timer.CH1.dont_capture_CCUVAL1	
	0	-	Timer.CH1.dont_capture_CCUVAL1	
	1	-	-	Timer.CH1.capture_CCUVAL1
			-	Timer.CH1.dont_capture_CCUVAL1
	2		-	Timer.CH1.capture_CCUVAL1
			-	Timer.CH1.dont_capture_CCUVAL1
	5		-	Timer.CH1.capture_CCUVAL1
1	-	-	Timer.CH1.compare_CCM1.CCU1	
2	-	-	Timer.CH1.compare_CCM2.CCU1	
3	-	-	Timer.CH1.compare_CCM3.CCU1	
4	-	-	Timer.CH1.compare_CCM4.CCU1	
5	-	-	Timer.CH1.compare_CCM5.CCU1	
6	-	-	Timer.CH1.compare_CCM6.CCU1	
7	-	-	Timer.CH1.disabled_compare_CCM7.CCU1	

Figure 7: Tabular overview (6) of the 2_channel_Timer_IP's operations encoding.

bf_CCM2_CH1_out	bf_CapIM2_CH1_out	TIM1_CCU2_ExtCap	default_interface_addr	Operation
0	3	1	-	Timer_CH1.capture_CCUVAL2
		0	-	Timer_CH1.dont.capture_CCUVAL2
	4	1	-	Timer_CH1.capture_CCUVAL2
			-	Timer_CH1.dont.capture_CCUVAL2
	6	-	Timer_CH1.dont.capture_CCUVAL2	
	7	-	Timer_CH1.dont.capture_CCUVAL2	
	0	-	Timer_CH1.dont.capture_CCUVAL2	
	1	-	-	Timer_CH1.capture_CCUVAL2
			-	Timer_CH1.dont.capture_CCUVAL2
	2		-	Timer_CH1.capture_CCUVAL2
			-	Timer_CH1.dont.capture_CCUVAL2
	5		-	Timer_CH1.capture_CCUVAL2
		-	Timer_CH1.dont.capture_CCUVAL2	
1	-	-	Timer_CH1.compare_CCM1.CCU2	
2	-	-	Timer_CH1.compare_CCM2.CCU2	
3	-	-	Timer_CH1.compare_CCM3.CCU2	
4	-	-	Timer_CH1.compare_CCM4.CCU2	
5	-	-	Timer_CH1.compare_CCM5.CCU2	
6	-	-	Timer_CH1.compare_CCM6.CCU2	
7	-	-	Timer_CH1.disabled.compare_CCM7.CCU2	

Figure 8: Tabular overview (7) of the 2_channel_Timer_IP's operations encoding.

bf_CCM3.CH1_out	bf_CapIM3.CH1_out	TIM1_CCU3.ExtCap	default_interface_addr	Operation
0	3	1	-	Timer.CH1.capture_CCUVAL3
		0	-	Timer.CH1.dont_capture_CCUVAL3
	4	1	-	Timer.CH1.capture_CCUVAL3
			-	Timer.CH1.dont_capture_CCUVAL3
	6	-	Timer.CH1.dont_capture_CCUVAL3	
	7	-	Timer.CH1.dont_capture_CCUVAL3	
	0	-	Timer.CH1.dont_capture_CCUVAL3	
	1	-	-	Timer.CH1.capture_CCUVAL3
			-	Timer.CH1.dont_capture_CCUVAL3
	2		-	Timer.CH1.capture_CCUVAL3
			-	Timer.CH1.dont_capture_CCUVAL3
	5		-	Timer.CH1.capture_CCUVAL3
1	-	-	Timer.CH1.compare_CCM1_CCU3	
2	-	-	Timer.CH1.compare_CCM2_CCU3	
3	-	-	Timer.CH1.compare_CCM3_CCU3	
4	-	-	Timer.CH1.compare_CCM4_CCU3	
5	-	-	Timer.CH1.compare_CCM5_CCU3	
6	-	-	Timer.CH1.compare_CCM6_CCU3	
7	-	-	Timer.CH1.disabled_compare_CCM7_CCU3	

Figure 9: Tabular overview (8) of the 2_channel_Timer_IP's operations encoding.

bf_CCM4_CH1_out	bf_CapIM4_CH1_out	TIM1_CCU4_ExtCap	default_interface_addr	Operation
0	3	1	-	Timer.CH1.capture_CCUVAL4
		0	-	Timer.CH1.dont.capture_CCUVAL4
	4	1	-	Timer.CH1.capture_CCUVAL4
			-	Timer.CH1.dont.capture_CCUVAL4
	6	-	Timer.CH1.dont.capture_CCUVAL4	
	7	-	Timer.CH1.dont.capture_CCUVAL4	
	0	-	Timer.CH1.dont.capture_CCUVAL4	
	1	-	-	Timer.CH1.capture_CCUVAL4
			-	Timer.CH1.dont.capture_CCUVAL4
	2		-	Timer.CH1.capture_CCUVAL4
			-	Timer.CH1.dont.capture_CCUVAL4
	5		-	Timer.CH1.capture_CCUVAL4
1	-	-	Timer.CH1.dont.capture_CCUVAL4	
2	-	-	Timer.CH1.compare_CCM1.CCU4	
3	-	-	Timer.CH1.compare_CCM2.CCU4	
4	-	-	Timer.CH1.compare_CCM3.CCU4	
5	-	-	Timer.CH1.compare_CCM4.CCU4	
6	-	-	Timer.CH1.compare_CCM5.CCU4	
7	-	-	Timer.CH1.compare_CCM6.CCU4	
	-	-	-	Timer.CH1.disabled.compare_CCM7.CCU4

Figure 10: Tabular overview (9) of the 2_channel_Timer_IP's operations encoding.

bf_CCM5.CH1_out	bf_CapIM5.CH1_out	TIM1.CCU5.ExtCap	default_interface_addr	Operation
0	3	1	-	Timer.CH1.capture_CCUVAL5
		0	-	Timer.CH1.dont_capture_CCUVAL5
	4	0	-	Timer.CH1.capture_CCUVAL5
		1	-	Timer.CH1.dont_capture_CCUVAL5
	6	Timer.CH1.dont_capture_CCUVAL5		
	7	Timer.CH1.dont_capture_CCUVAL5		
	0	Timer.CH1.dont_capture_CCUVAL5		
	1	-	-	Timer.CH1.capture_CCUVAL5
			-	Timer.CH1.dont_capture_CCUVAL5
	2		-	Timer.CH1.capture_CCUVAL5
			-	Timer.CH1.dont_capture_CCUVAL5
	5		-	Timer.CH1.capture_CCUVAL5
			-	Timer.CH1.dont_capture_CCUVAL5
1	-	-	Timer.CH1.compare_CCM1.CCU5	
2	-	-	Timer.CH1.compare_CCM2.CCU5	
3	-	-	Timer.CH1.compare_CCM3.CCU5	
4	-	-	Timer.CH1.compare_CCM4.CCU5	
5	-	-	Timer.CH1.compare_CCM5.CCU5	
6	-	-	Timer.CH1.compare_CCM6.CCU5	
7	-	-	Timer.CH1.disabled_compare_CCM7.CCU5	

Figure 11: Tabular overview (10) of the 2_channel_Timer_IP's operations encoding.

TIM0_OvfIntRes	bf_OvfIntEn.CH0_out	bf_ACTVAL.CH0_flipflop_out	bf_ResIM.CH0_out	Operation
1	-	-	-	Timer.CH0.overflow_int_reset
0	1	0	7	Timer.CH0.overflow_int_set

Figure 12: Tabular overview (11) of the 2_channel_Timer_IP's operations encoding.

TIM1_OvfIntRes	bf_OvfIntEn.CH1_out	bf_ACTVAL.CH1_flipflop_out	bf_ResIM.CH1_out	Operation
1	-	-	-	Timer.CH1.overflow_int_reset
0	1	0	7	Timer.CH1.overflow_int_set

Figure 13: Tabular overview (12) of the 2_channel_Timer_IP's operations encoding.

8 2_channel_Timer_IP Traces

8.1 sw_write_non_virtual_bf_TimEn_CH0_Trace Trace

The `sw_write_non_virtual_bf_TimEn_CH0_Trace` has one length-one entry transition and one final state. The trigger is `wr.en=1` with `address=0x00`; the final action loads `TimEn` from previous-cycle write-data bit 0. It describes a clocked write, not a persistent enable-control FSM or an explicit idle state. The same `CTRLSTAT` word write updates the other control fields through parallel traces. `TimEn` controls counter hardware updates only.

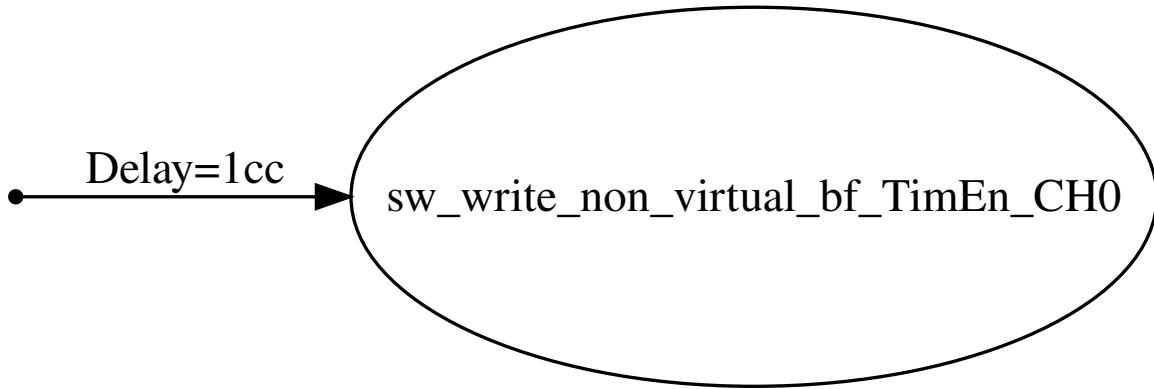


Figure 14: Trace visualization as state chart.

8.2 hw_write_virtual_bf_ACTVAL_CH0_Trace Trace

The `hw_write_virtual_bf_ACTVAL_CH0_Trace` is the hardware-update case with `wr.en=0` and `bf_ACTVAL_CH0_peripheral`. Its length-one entry transition reaches a single final state that copies previous-cycle `bf_ACTVAL_CH0_flipflop.in` into `ACTVAL` storage. A separate `_1_Trace` covers `wr.en=1` with hardware write enabled, consistent with hardware priority. No explicit idle state is present. This trace consumes selected-input and enable helpers whose complete combinational definitions are absent from the XML; in the RTL they select decrement or reload data.

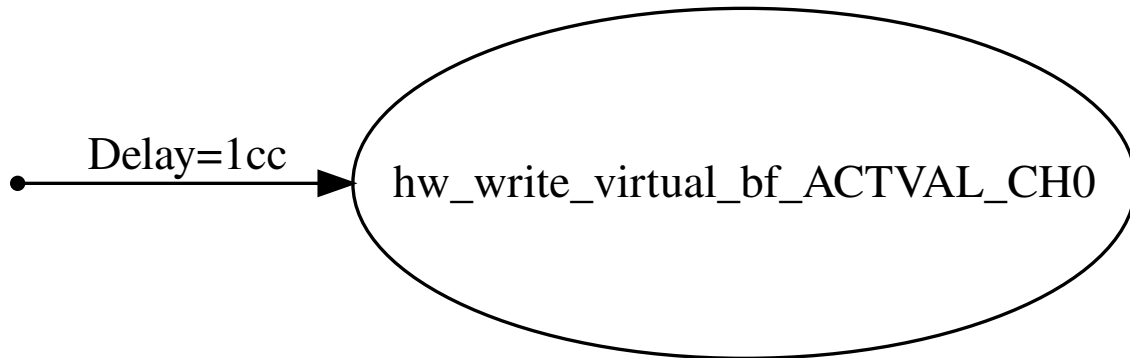


Figure 15: Trace visualization as state chart.

8.3 read_CH0_CTRLSTAT_Trace Trace

The `read_CH0_CTRLSTAT_Trace` has a zero-length entry transition to one final state. This particular trace requires `wr.en=1`, `rd.en=1`, and `address=0x00`; the separate `read_CH0_CTRLSTAT_1_Trace` cov-

ers `wr_en=0`. The final action returns the concatenated `TimEn`, `ResIM`, `CntIM`, and `OvfIntEn` fields, with unused upper bits zero in the RTL. Thus it specifies a combinational read during a simultaneous write request, not a multi-cycle read protocol or a required explicit idle state. The interrupt output is not part of `CTRLSTAT` readback.

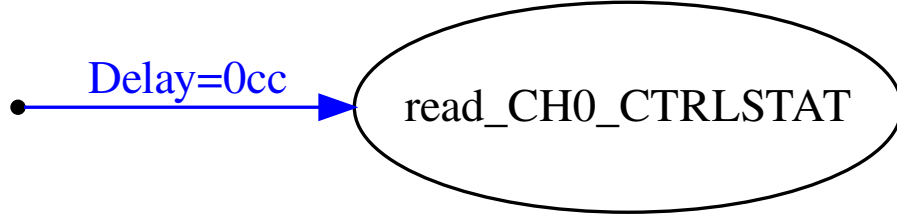


Figure 16: Trace visualization as state chart.

8.4 Timer_CH0_dont_count_Trace Trace

The `Timer_CH0_dont_count_Trace` begins with an unconditional length-two entry into its intermediate state, then a length-one transition conditioned on `TimEn=0` and `default_interface_addr!=0x04`. The final action retains previous-cycle `ACTVAL`. This is one disabled-counter hold case, not every inactive-count condition; other numbered traces cover additional cases. The address exclusion is present even when `wr_en=0`. These are formal observation/history stages, not extra hardware pipeline registers, and they do not prevent a valid software `ACTVAL` write.

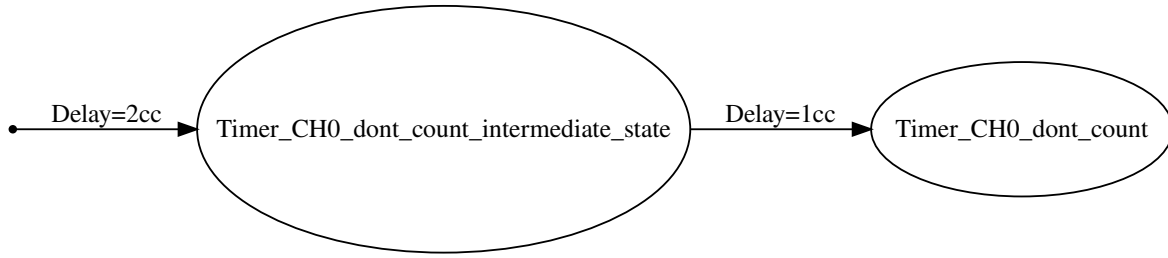


Figure 17: Trace visualization as state chart.

8.5 Timer_CH0_reset_2_maxval_Trace Trace

The `Timer_CH0_reset_2_maxval_Trace` describes the automatic-reload subset. An unconditional length-two entry reaches the intermediate state; a length-one transition then requires `TimEn=1`, `ACTVAL=0`, and `ResIM=7`. The final action sets `ACTVAL` to previous-cycle `MAXVAL`. External-reload cases are covered by other numbered traces. The two preparatory cycles belong to the formal trace structure, not a hardware requirement to spend three cycles reloading. This is neither global reset nor an up-count operation, and it leaves `MAXVAL` and other registers unchanged.

8.6 Timer_CH0_down_count_Trace Trace

The `Timer_CH0_down_count_Trace` has an unconditional length-two entry followed by a length-one conditional transition. This particular guard requires `TimEn=1`, `ResIM=3`, previous-cycle `TIM0_ExtRes=0`, `CntIM=3`, and previous-cycle `TIM0_ExtCnt=1`. It also excludes `ACTVAL` values 0 through 7. The final action decrements previous-cycle `ACTVAL` by one. Numbered companion traces cover other input modes. The exclusion of 1 through 7 is a model coverage defect: the RTL requires only `ACTVAL!=0` for decrementing. The intermediate state supports formal sampled-history evaluation, not a separate

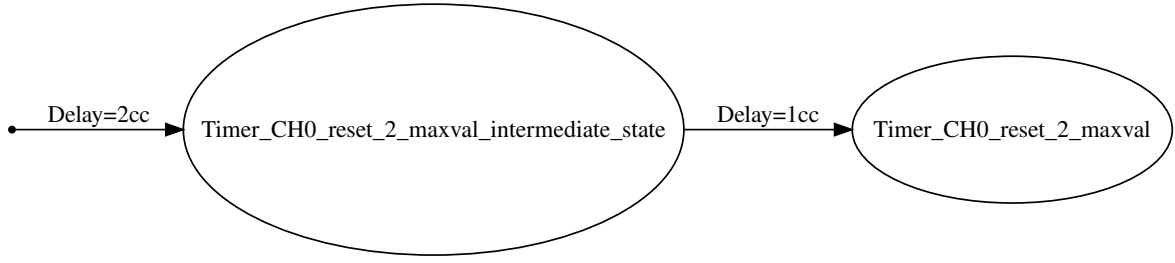


Figure 18: Trace visualization as state chart.

counter operating mode or a hardware prescaler.

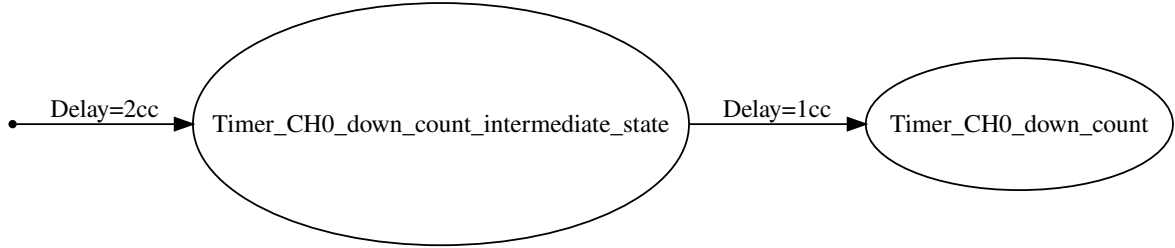


Figure 19: Trace visualization as state chart.

8.7 Timer_CH0_capture_CCUVAL0_Trace Trace

The `Timer_CH0_capture_CCUVAL0_Trace` is the high-level capture case: after an unconditional length-two entry, a length-one transition requires `CCM0=0`, `CapIM0=3`, and previous-cycle `TIM0_CCU0_ExtCap=1`. The final actions copy previous-cycle `ACTVAL` into `CCUVAL0` and constrain previous-cycle `ExtComp` to zero. `TimEn` is not tested. Numbered companion traces cover other capture-input modes. The capture is clocked and can repeat while the sampled input stays high; it does not record the exact asynchronous event time. The intermediate state is part of formal history handling, not an additional capture buffer.

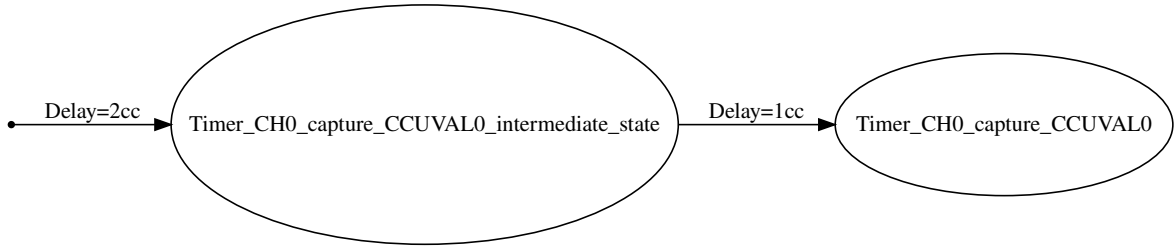


Figure 20: Trace visualization as state chart.

8.8 Timer_CH0_dont_capture_CCUVAL0_Trace Trace

The `Timer_CH0_dont_capture_CCUVAL0_Trace` covers one inactive high-level capture case. Its unconditional length-two entry is followed by a length-one transition requiring `CCM0=0`, `CapIM0=3`, previous-cycle `ExtCap=0`, and `address!=0x10`. The final actions retain previous-cycle `CCUVAL0` and constrain previous-cycle `ExtComp` to zero. The address exclusion avoids a potential software write even if `wr_en` is low; this is not a universal no-capture rule. `CCUVAL0` is the capture/compare storage,

while ACTVAL is the counter. The RTL still accepts a software write when no hardware capture is active.

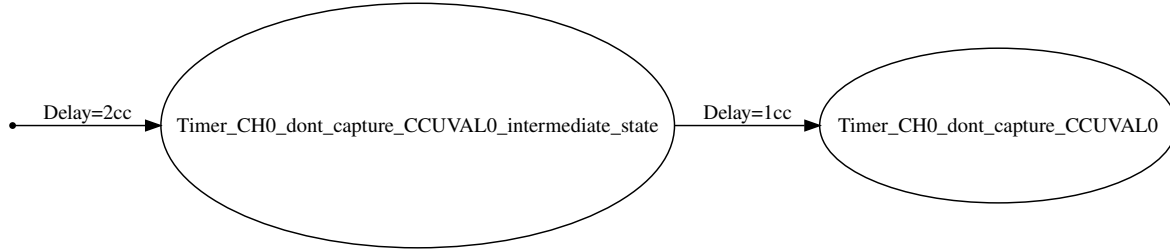


Figure 21: Trace visualization as state chart.

8.9 Timer_CH0_compare_CCM1_CCU0_Trace Trace

The Timer_CH0_compare_CCM1_CCU0_Trace has a zero-length entry triggered solely by CCM0=1. Its single final state drives ExtComp to the unsigned relation ACTVAL less than CCUVAL0, including both the true and false results. Entry does not require the less-than relation itself to be true. This models a combinational comparison, independent of TimEn, not a latched compare-event FSM.

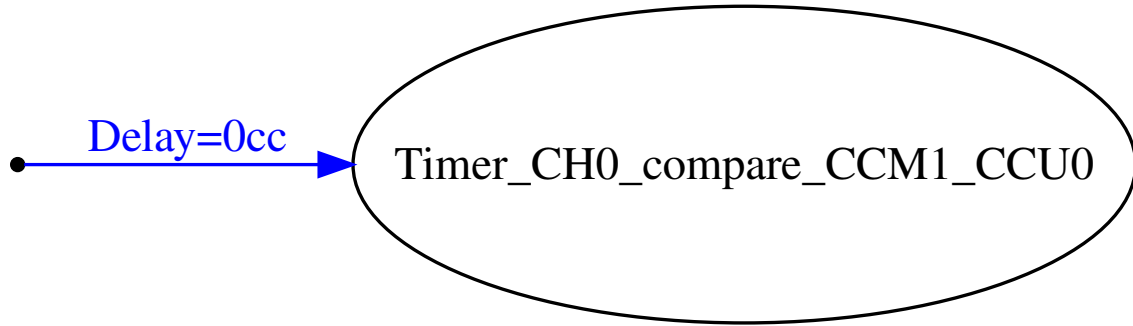


Figure 22: Trace visualization as state chart.

8.10 Timer_CH0_compare_CCM5_CCU0_Trace Trace

The Timer_CH0_compare_CCM5_CCU0_Trace has a zero-length entry triggered solely by CCM0=5. Its single final state drives ExtComp to the equality result between ACTVAL and CCUVAL0, including zero when unequal. CCM5 denotes mode encoding 5, not a register containing the reference value. The trace applies while the timer is enabled or disabled and does not imply a one-clock pulse or that counting reached a threshold.

8.11 Timer_CH0_disabled_compare_CCM7_CCU0_Trace Trace

The Timer_CH0_disabled_compare_CCM7_CCU0_Trace has a zero-length entry triggered by CCM0=7 and a single final action ExtComp=0. It does not require TimEn=0 and does not disable the channel counter. The RTL also disables hardware capture in mode 7 while preserving software access to CCUVAL0. The trace describes an output constraint, not execution of a compare operation in a disabled channel.

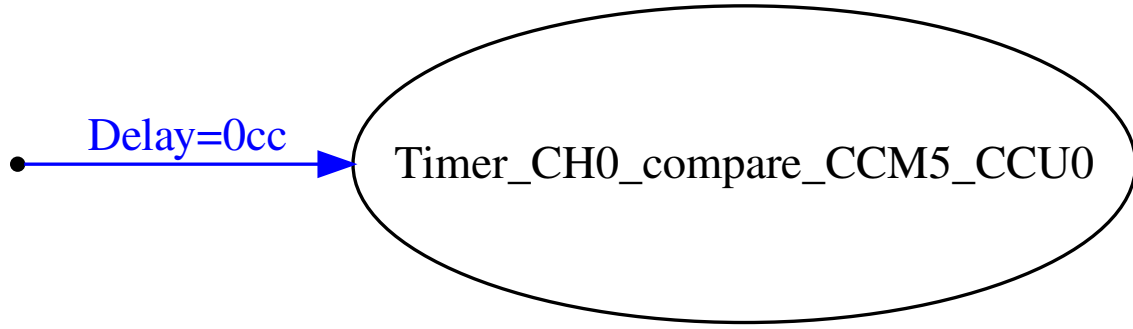


Figure 23: Trace visualization as state chart.

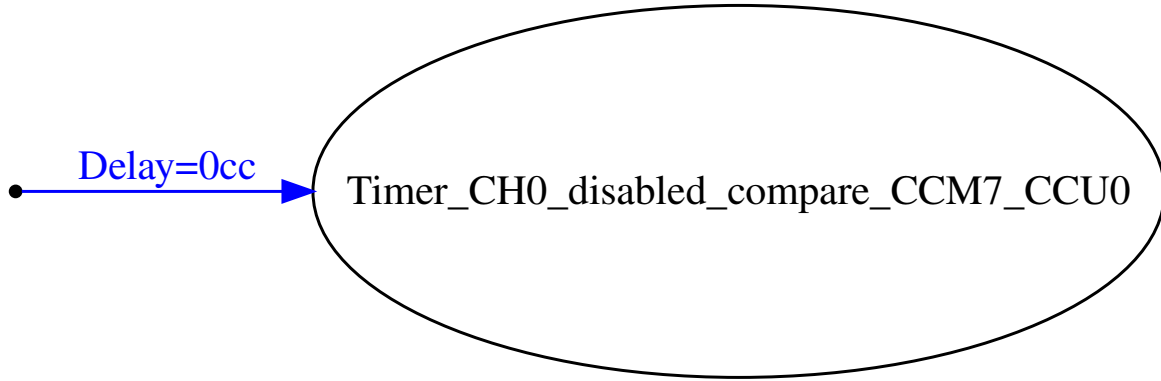


Figure 24: Trace visualization as state chart.

8.12 Timer_CH0_overflow_int_reset_Trace Trace

The `Timer_CH0_overflow_int_reset_Trace` has an unconditional length-two entry into an intermediate state, followed by a length-one transition requiring `TIM0_OvfIntRes=1`. Its final action clears `TIM0_OvfInt`. The RTL clear is a synchronous input action with priority over the interrupt condition, not a software register write or a three-cycle hardware handshake. Other RTL deassertion causes, including the zero-condition disappearing, are not covered by this trace.

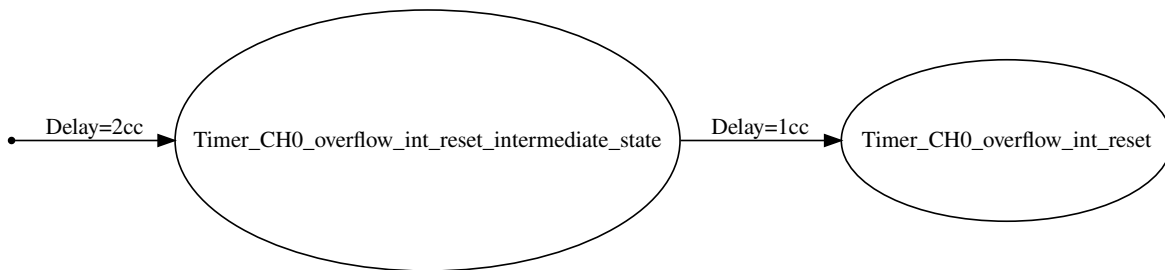


Figure 25: Trace visualization as state chart.

8.13 Timer_CH0_overflow_int_set_Trace Trace

The `Timer_CH0_overflow_int_set_Trace` has a length-one entry requiring `TIM0_OvfIntRes=0`, `OvfIntEn=1`, `ACTVAL=0`, and `ResIM=7`, followed by a single final state assigning `TIM0_OvfInt=1`. There is no `TimEn` condition or explicit idle state. This models a registered zero-condition indication, not maximum-value overflow or manual software assertion. The RTL permits any nonzero `ResIM` and

recomputes the output each clock; this trace covers only mode 7 assertion, not complete interrupt behavior.

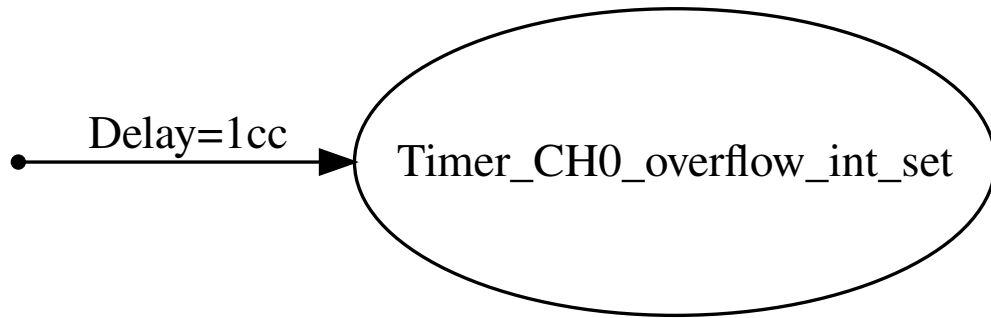


Figure 26: Trace visualization as state chart.